

ANEESH RAJ DONTI

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WORK EXPERIENCE

AI Engineer— Tata Electronics Pvt. Ltd.

Nov 2025 –present

- Supported **data-driven production scheduling and capacity optimization** using **4D spatiotemporal simulations** in **NVIDIA Omniverse**, modeling **10+ manufacturing scenarios** for a large-scale semiconductor facility
- Performed **clash detection, bottleneck analysis, and throughput forecasting**, helping identify **15–20% potential improvements** in resource utilization and production flow across critical workstations
- Conducted **scenario-based simulations and KPI analysis** (utilization, cycle time, throughput), reducing planning uncertainty and improving **capacity planning accuracy by 12–18%** for the Dholera semiconductor plant

R&D Intern — NIT Warangal

Nov 2023 – Dec 2023

- Designed and enhanced image processing pipelines using **Local Binary Patterns (LBP)**, **OpenCV**, and **NumPy**, achieving a measurable runtime gain of **17%**.
- Leveraged **TensorFlow**, **PyTorch**, and **Keras** to prototype **CNN-based models** with transfer learning, improving classification performance.
- Curated benchmark datasets (**Corel-250**), automated evaluation with ranking models (**k-NN**, **SVM**), and standardized documentation for reproducibility.

EDUCATION

National Institute of Technology (NIT), Warangal,

Dec 2021 – Apr 2025

B.Tech in Mechanical Engineering

Coursework: Computer Programming, Matlab, Python, Data Structures and Algorithms, Management Science, Artificial Intelligence In CyberPhysical Systems

TECHNICAL SKILLS

Languages & Analysis: Java,Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL (Joins, Window Functions, Optimization)

Data Analytics & BI Tools:Tableau, Power BI, IBM Cognos, Excel (Pivot Tables, Power Query, Macros)

Cloud & Tools: AWS (EC2, S3, RDS, IAM, Lambda, CloudWatch, API Gateway), Git, GitHub, Docker, CI/CD pipelines, Postman, VS Code, Figma.

PROJECTS

Real-Time Steel Damage Classification with AWS Cloud Technology | *AWS Rekognition, Lambda, API Gateway, S3*

- Built a **serverless defect detection system** on AWS (Rekognition, Lambda, API Gateway, S3) capable of processing **5,000+ images/day** for real-time quality control.
- Optimized ML models, **improving defect classification accuracy** from **82% to 93%**, reducing false negatives in automotive and aerospace inspections.
- Leveraged **parallel processing and cloud integration**, cutting analysis latency by **25%** and enabling scalable deployment across multiple production units.

Domino's Branch Sales Analytics and Visualization | *MS SQL Server, Power BI*

- Built a **sales analytics platform** in **MS SQL Server** to manage and query **50,000+ transaction records**, improving reporting speed by **35%**.
- Designed a **Power BI dashboard** with advanced **data modeling and DAX**, providing real-time insights into **10+ KPIs** such as sales performance, customer retention, and product mix.
- Conducted **trend analysis and demand forecasting**, boosting forecast accuracy by **20%** and enabling data-driven operational planning during peak hours.

NewsPress – Personalized News Recommender System | *HuggingFace Transformers (MiniLM), BERT*

- Built a **BERT-based recommender system** using **MiniLM embeddings, cosine similarity, and content-based filtering** on the Microsoft MIND dataset to deliver **personalized top-N news recommendations**.
- Modeled **user behavior profiles** from click history, performed **semantic similarity ranking**, and evaluated performance using **accuracy vs clicks analysis**
- Developed and deployed an **end-to-end NLP pipeline** with **Python, Pandas, Flask, interactive data visualizations (Chart.js)**, and cloud deployment

Leadership & Organizational Experience

- **General Secretary — NSS, NIT Warangal (2023)** – Directed 6+ large-scale community initiatives, including a Blood Donation Camp (350+ units) and Unity Run (500 participants). Spearheaded awareness drives, women's self-defense workshops, and technical talks, engaging 1,200+ students and improving volunteer participation by 35%.
- **Head of Logistics — SpringSpree, NIT Warangal (2023)** – Managed logistics operations for South India's 2nd-largest cultural fest (5,000+ attendees), coordinating 100+ volunteers and 20+ vendors. Perfected scheduling and resource allocation, cutting setup time by 2 hours/day and ensuring seamless event execution.